

Bill of Materials (PDF release)

VBF-T8R1NOFORCE-BOM-01 | case t8_r1_noforce | 2026-08-25 | dual caliber g/cell + kg/kWh on 19.4147 Wh (1C contract)

Component	Mass (g/cell)	Mass (kg/kWh)	In contract mass	Source / basis
Positive active material (NMC811)	18.1870	0.9368	YES	r6_v9_energy.json layer_kg_m2 x area
Negative active material (graphite)	11.2793	0.5810	YES	r6_v9_energy.json layer_kg_m2 x area
Positive current collector (Al, 8 um)	2.6550	0.1368	YES	r6_v9_energy.json layer_kg_m2 x area
Negative current collector (Cu, 6 um)	6.6079	0.3404	YES	r6_v9_energy.json layer_kg_m2 x area
Separator (10 um)	0.2586	0.0133	YES	r6_v9_energy.json layer_kg_m2 x area
CONTRACT TOTAL (electrolyte excluded)	38.9878	2.0082	YES	r6_v9_energy.json mass_kg = 0.0389878 kg
Electrolyte (1 M LiPF6 EC:DMC)	9.5520	0.4920	no	pore volume 7.960e-6 m3 x 1200 kg/m3; density = literature value; fill factor 1.0
Positive binder (PVDF)	0.3637	0.0187	no	2 wt% of active mass - literature default (excluded from contract mass)
Positive conductive carbon	0.3637	0.0187	no	2 wt% of active mass - literature default (excluded from contract mass)
Negative binder (CMC+SBR)	0.2256	0.0116	no	2 wt% of active mass - literature default (excluded from contract mass)
Negative conductive carbon	0.1128	0.0058	no	1 wt% of active mass - literature default (excluded from contract mass)
Enclosure, tabs, termination	0.0000	0.0000	no	not modeled in this virtual case
GRAND TOTAL (incl. electrolyte + binder/additive)	49.6057	2.5551	no	annotated total

Contract mass = 38.99 g (criterion <= 40 g, PASS). Electrolyte, binder and additives are listed supplementary: electrolyte density 1.2 g/cm3 and binder/additive fractions (2/2 wt% positive, 2/1 wt% negative) are literature defaults, recorded as annotations, excluded from the calc-energy contract caliber. Enclosure/tabs not modeled.